

MD. MAJIDUL ISLAM

AI & ML · GENAI · AGENTIC AI

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“Building intelligent systems that turn data into decisions and automation into advantage”

Work Experience

TulipTech Limited

Dhaka, Bangladesh

AI/ML ENGINEER

Aug. 2025 - Present

- **Built and optimized Edge AI pipelines** on NVIDIA Jetson platforms for low latency inference using TensorRT, CUDA acceleration and FP16/INT8 optimizations
- **Developed multimodal AI systems** combining vision, text and document understanding including OCR pipelines for structured and unstructured data
- **Implemented Retrieval Augmented Generation (RAG)** using LangChain and CrewAI with vector databases for reliable knowledge grounding
- **Designed and orchestrated multi-agent AI systems** using CrewAI and LangGraph for task decomposition, tool usage and collaborative workflows
- **Fine-tuned and evaluated large language models** on domain-specific datasets using in-house GPU clusters, optimizing accuracy
- **Implemented LLM observability and hallucination analysis** using Phoenix and Langfuse for prompt tracing, evaluation, latency and cost monitoring

Synova Information Systems Limited

Dhaka, Bangladesh

AI PROGRAMMER DATA

Mar. 2025 - Aug. 2025

- **Data Driven AI Solutions:** Built and optimized data driven AI solution with backend
- **AI Backend Development:** Create and Implement FastAPI backend for AI tools
- **Agentic AI Solution:** Designed multi-agent AI systems to perform autonomous reasoning and decision-making tasks
- **Big Data Processing:** Processed large datasets using tools like Pandas, PySpark for real time insights
- **MLOps Implementation:** Set up automated pipelines for model training, testing, deployment and monitoring using tools like MLflow, LlamaIndex
- **LLM Oriented Automation:** Automated reasoning, task execution, and content generation using large language models
- **Multi Agent Framework Integration:** Leveraged frameworks like LangChain, CrewAI and AutoGen to orchestrate collaboration between AI agents

TulipTech Limited

Dhaka, Bangladesh

SOFTWARE ENGINEER(AI)

Oct. 2022 - Mar. 2025

- **Designed and Deployed AI Solutions** using Python and FastAPI following MLOps best practices for scalable and maintainable machine learning workflows
- **Implemented End to End ML Architecture** for training, testing, deployment and monitoring, ensuring robust model life cycle management
- **Led Cloud & DevOps Integration** with Microsoft Azure to provision, monitor and optimize cloud-based AI infrastructure
- **Engineered FAST APIs and Data Architectures** maintaining performant, secure APIs and managing document based databases (MongoDB)
- **Contributed to Project Management** supporting agile delivery through cross-team coordination and CI/CD pipeline maintenance
- **Multi Agent Framework Integration:** Leveraged frameworks like LangChain, CrewAI and AutoGen to orchestrate collaboration between AI agents

Bangladesh Navy

Bangladesh Naval Academy,
Bangladesh

OFFICER CADET

Dec. 2016 - Mar. 2018

- **Leadership & Discipline:** Completed intensive military training, strengthening leadership, teamwork and decision-making under pressure
- **Maritime & Tactical Training:** Acquired hands on experience in naval operations, navigation and tactical combat strategies
- **Strategic Planning & Crisis Management:** Trained to operate effectively in high-stress environments, making quick and strategic decisions
- **Technical & Weaponry Knowledge:** Gained foundational knowledge in naval engineering, weapon systems and maritime technologies
- **Ethics & Professionalism:** Maintained strong discipline, integrity and commitment to service throughout military service

Projects

Author & Maintainer | PyPI: [personakit](#) | GitHub: [Majidul17068/personakit](#)

Independent OSS | MIT License

PERSONAKIT — DECLARATIVE AGENT BUILDER FOR LLMs (OPEN SOURCE)

2026 – Present

- **Designed and shipped** an open-source Python library that builds role-based LLM agents declaratively – persona, frameworks, probes, red flags and recommendation themes as a single data object instead of hand-wired chains.
- **Production wiring:** streaming (`analyze_stream`), OpenTelemetry hooks (3 span points), token cost tracking across 30 models, multi-turn conversational sessions, and a multi-turn tool-calling loop with cross-provider format normalization (OpenAI, Anthropic, plus 100+ providers via LiteLLM).
- **Differentiator:** red flags as a first-class primitive – trigger → severity → action → citation, matched both deterministically (regex) and semantically (LLM); audit-grade output for clinical, legal, fintech and compliance use cases.
- **Quality & reach:** 93 unit tests, `mypy --strict` clean across 29 source files, 7 bundled domain specialists (engineering, fintech, support, scrum, legal, clinical, education), 11 releases shipped to PyPI, full GitHub Actions CI + Trusted Publishing pipeline.

Live in Production | Multi-Agent RAG | Care & Medication

Healthcare AI

EMPATHIKA — AI PLATFORM FOR HEALTHCARE & RESEARCH

2026 – Present

- **Currently live in production** – designed and shipped a healthcare AI platform serving real users for care planning, medication safety and clinical research workflows.
- Multi-agent system combining Retrieval-Augmented Generation over internal clinical records with external medical knowledge for grounded, citation-backed responses.
- Real-world deployment of personakit – the open-source library drives the clinical agents (persona, frameworks, red flags) declaratively in production.

Live in Production | Agentic RAG for Large Organizations

Kazi Farms Group

HR POLICY CHATBOT – EMPLOYEE QUERY ASSISTANT

2025

- **Live in production** at a large enterprise – thousands of employees query internal HR and policy documents in natural language with full source attribution.
- Agentic RAG built on DeepSeek models and ChromaDB vector database; supports both English and Bangla with citation-backed responses.

Live in Production | Smart Financial Workflow Automation

Enterprise Client

BANK RECONCILIATION – AI-POWERED ENTERPRISE FINANCE

2025

- **Live in production** at a large enterprise – AI system that automatically matches bank statements with internal financial records.
- Reduces manual finance work, improves accuracy and accelerates monthly close – making reconciliation intelligent and effortless at scale.

Sales forecasting (Big Data)

Dhaka, Bangladesh

AI SOLUTION – SALES FORECASTING

2025

- Developed a machine learning based sales forecasting system leveraging historical sales data, seasonal trends and external factors.
- Enabled data driven decision making for marketing, inventory and supply chain teams through predictive analytics dashboards.

Healthcare Data Automation Project

TulipTech Limited

AI AUTOMATION – NHS MEDICINE PARSER

2024

- Developed an automated system to parse and extract critical data from NHS medical records, improving healthcare data accessibility.

Drug Database Processing System

TulipTech Limited

AI DEVELOPMENT – DMD+D MEDICINE AI PROCESSOR

2024

- Created an AI based engine for analyzing drug metadata from DMD+D to enhance pharmaceutical data workflows and accuracy.

Care Home Personalized AI Healthcare System

TulipTech Limited

AI SOLUTION ARCHITECT & DEVELOPMENT – CARE HOME PLAN & AI AGENTS

2024

- Designed an AI driven care plan system with intelligent agents
- support personalized monitoring and recommendations for care home residents

Research

Multi Factor Authentication Kerberos

IEEE SEM Vol -12, 2024

INTERNET BANKING CYBER SECURITY IN BANGLADESH

Jan. 2024

- Multi factor authentication solutions for Bangladeshi Banks

Technical Skills

EDGE AI & GPU-ACCELERATED SYSTEMS

- Edge Platforms: NVIDIA Jetson
- Parallel Computing: CUDA, GPU memory hierarchy & optimization
- Model Deployment: TensorRT, ONNX Runtime, PyTorch, TensorFlow
- Performance Optimization: kernel fusion, batching, FP16/INT8 quantization
- Real-Time Inference: low-latency vision and sensor processing pipelines

LLM SYSTEMS & MULTI-AGENT ARCHITECTURES

- LLM Frameworks: Hugging Face Transformers, LangChain, LlamaIndex
- RAG Systems: retrieval design, embedding analysis, vector stores (ChromaDB, Pinecone etc)
- Multi-Agent Systems: CrewAI, LangGraph, task/tool-based agent orchestration
- LLM Ops & Evaluation: Phoenix, Langfuse (tracing, evaluation, cost/latency analysis)
- LLM APIs: OpenAI, DeepSeek, Groq, Ollama

ML INFRASTRUCTURE & PRODUCTION SYSTEMS

- Backend Services: FastAPI, Streamlit
- MLOps: MLflow, model versioning, experiment tracking
- CI/CD & DevOps: Docker, GitHub Actions, Jenkins
- Cloud Platforms: AWS, GCP, Azure

DATA & APPLIED MACHINE LEARNING

- Pandas, NumPy, Scikit-Learn

Education

Southeast University, Bangladesh

BSC. IN COMPUTER SCIENCE AND ENGINEERING

- CGPA: 3.66 / 4.00

Dhaka, Bangladesh

Oct. 2018 – Oct. 2022

Adamjee Cantonment College

HIGHER SECONDARY CERTIFICATE

- GPA: 5.00 / 5.00

Dhaka, Bangladesh

Aug. 2016

Muslim Modern Academy

SECONDARY SCHOOL CERTIFICATE

- GPA: 5.00 / 5.00

Dhaka, Bangladesh

May. 2014

Memberships, Training and Certifications

PROFESSIONAL MEMBERSHIPS

- 2026 – Present
- Member, Project Management Institute (PMI)

Global

PROFESSIONAL CERTIFICATIONS

- 2025
- Oracle Certified Foundation Associate, Certified AI Foundation
- 2024
- Agile Project Management Practitioner, Project Management
- 2023
- Admin Official Certification, Project Management
- 2022
- Cyber Security (Ethical Hacking), Arena Web Security

Oracle University
HP LIFE
monday.com
Dhaka, Bangladesh

TRAINING

- 2023
- Industrial Data Science and Machine Learning with AI, HR Ventures
- 2021
- Ethical Cracking and Hacking, SkillNation Bangladesh

Dhaka, Bangladesh
Dhaka, Bangladesh

Extracurricular Activity

AI/ML Professional Community | LinkedIn Group

MENTOR

- Mentor practitioners on production AI/ML topics: LLM systems, RAG pipelines, multi-agent architectures, MLOps, edge deployment and career growth.
- Share open-source work and engineering writeups with the community, supporting members building real-world AI products.

Online | Global
2025 – Present

96.4 Spice Fm

RADIO JOCKEY

- Created content, wrote scripts and conducted live interviews with artists, entrepreneurs and public figures to drive audience engagement.
- Collaborated with producers and broadcast team to ensure smooth broadcasting, while maintaining a lively and energetic on air presence

Dhaka, Bangladesh
Oct. 2020 - Oct. 2022